

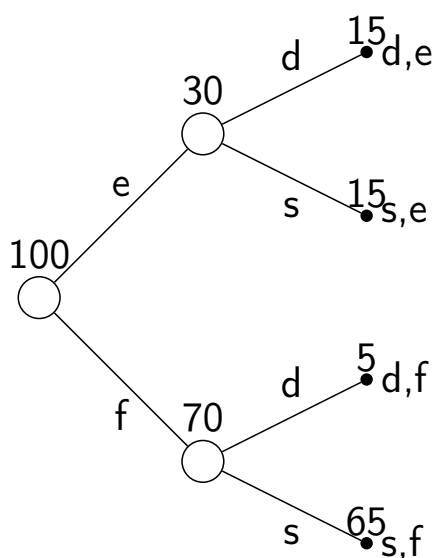
Lesson 1.1: Wald's Analysis

Introduction to Decision-making

Victor Jauregui

September 12, 2022

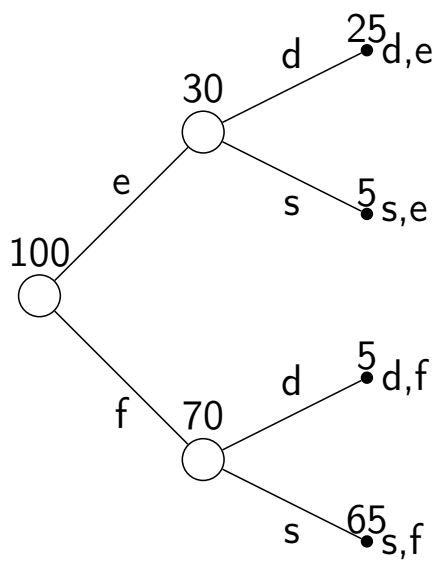
Mission decision analysis



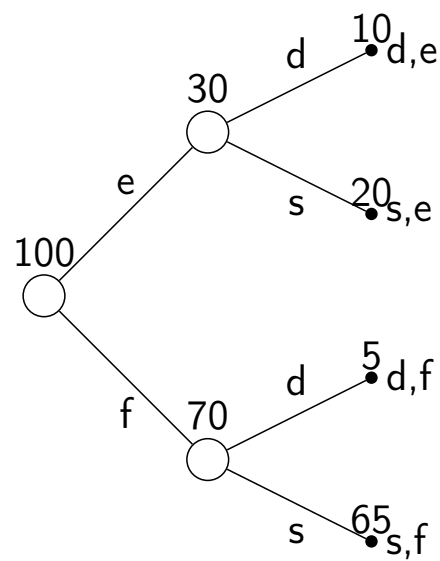
- ▶ less chance of being hit in engines (e) than fuselage (f)
- ▶ higher chance of survival (s) for hits in fuselage than engine
- ▶ armouring fuselage less likely to affect survival
- ▶ note over-representation of hits to the fuselage (lower branch) in total outcomes which survive

Survival rate: 80% (15 + 65)

Comparison

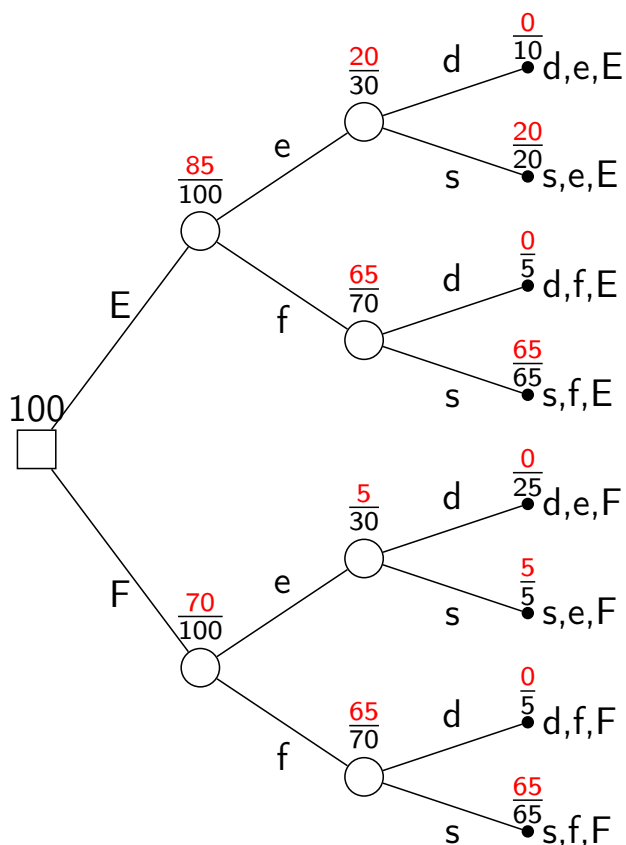


Survival rate: 70% (5 + 65)



Survival rate: 85% (20 + 65)

Full decision analysis



red bombers which
would survive
mission

- Example:
If we were to choose to protect the engines (branch E), then 20 (out of 30) bombers hit in the engines would survive